

Income Inequality, Capitalism, And Diversity: a Goodness-of-fit Decomposition Perspective

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Abstract

In this research, we extend the analysis of a merged dataset from two previous studies conducted by Ashraf and Galor (AER, 2013) and Sturm and De Haan (AER, 2015), conducted in 2022 by Ibrahim Ben Araar.

The objective of our study is to enhance the understanding of the dataset by employing the rego method, which allows for the decomposition of the R-squared goodness-of-fit measure of an ordinary least squares (OLS) regression model into contributions from individual regressors.

Through this decomposition, we aim to gain insights into the specific contributions of different regressors and their impact on the overall model fit.

Income Inequality and Its Significance for Research

- Income inequality is a pressing issue that affects individuals and societies worldwide, leading to social and economic disparities.
- Researching income inequality helps us understand its causes, consequences, and potential solutions, enabling policymakers to implement effective strategies for promoting a more equitable and inclusive society.

Previous Research

Desmet, K., Ortuño-Ortín, I., & Weber, S., 2009

- Linguistic diversity demonstrates both statistical and economic significance on redistribution.
- Linguistic distances provide a more accurate understanding of the relationship.

Impact on Redistribution:

- Average redistribution level of 9.5% of GDP in the dataset.
- Increase in linguistic diversity (by one standard deviation) leads to a decrease of approximately one percentage point in redistribution.

Jan-Egbert Sturm & Jakob de Haan, 2015

- There is **no consistent relationship** between **economic freedom** (proxied by an adjusted economic freedom index) and **income inequality** (proxied by Gini coefficients based on gross income) across a large sample of countries.
- The **impact** of **ethno-linguistic fractionalization** on **income redistribution** is **contingent** upon the level of economic freedom.
- Countries** with **high fractionalization** tend to have **less** income redistribution, while **capitalist countries with low fractionalization** exhibit **more** income distribution.

Ashraf QH & Galor O, 2013

This research explores the impact of ancient migratory patterns on economic development.

It finds that genetic diversity resulting from the exodus of Homo sapiens out of Africa has influenced comparative economic development,

- with **intermediate levels** of genetic diversity **benefiting** Asian and European populations,
- while **high diversity** in Africa and **low diversity** in Native American populations have **hindered** development.

Ibrahim Ben Araar, 2023

The research shows a **negative linear effect** of genetic diversity on redistribution, contradicting the proposed quadratic relationship.

The impact of genetic diversity on redistribution **is influenced** by the level of economic freedom.

Ethnic fractionalization also has a **negative effect** on redistribution.

The study supports the importance of a **capitalist system and genetic diversity** for **higher redistribution levels**. However, the suitability of genetic diversity as a variable is questioned.

Contribution to Research

An extension of the analysis of a merged dataset of Ashraf and Galor (AER, 2013) and Sturm and De Haan (AER, 2015) by Ibrahim Ben Araar in 2022 with the help of **decomposition of the R-squared goodness-of-fit measure** of an OLS regression model **into contributions of individual regressors by rego method**.

Goodness-of-fit

Goodness-of-fit tests are used to evaluate the fit of regression models and to determine **how well the model explains the variation in the dependent variable.**

It is a measure of **how closely the observed data matches** the expected data under the model in question.

R-squared Goodness-of-fit Measure

It is also known as the coefficient of determination.

R-squared measures the strength of the relationship between the model and the dependent variable on a scale of 0 to 100%.

- ✓ A **higher R-squared value** indicates that the independent variable(s) in the model **explain a large proportion of the variation** in the dependent variable;
- ✓ while a **lower R-squared value** indicates that the independent variable(s) in the model **explain a small proportion of the variation** in the dependent variable.

Rego Method

Rego is a Stata module that can be used to **decompose the R-squared goodness-of-fit measure** of an OLS regression model into **contributions of individual regressors** according to the Shapley and Owen values.

- This method is useful for grouping variables together if they represent similar ideas.
- The rego module is based on the covariance matrix of the data for fast computation.

Rego is written and maintained by **Frank Huettnner** and **Marco Sunder**, University of Leipzig.

How Decomposition Works

The decomposition works by calculating the contribution of each independent variable to the R-squared value of the model.

- ✓ The **Shapley value method** calculates the average contribution of each independent variable **across all possible combinations** of variables;
- ✓ while the **Owen value method** calculates the contribution of each independent variable **based on its membership in pre-defined groups**.

Shapley Value Method

The Shapley value method is a solution concept in cooperative game theory that is used to distribute the total gains of a game fairly among players. It assigns a unique distribution of a total surplus generated by the coalition of all players.

The Shapley value is characterized by a collection of desirable properties, including:

- Efficiency: The sum of the Shapley values of all players equals the total surplus generated by the coalition of all players.
- Symmetry: If two players have the same marginal contribution to all possible coalitions, then they have the same Shapley value.
- Additivity: The Shapley value of a player in a game that is the combination of two sub-games is the sum of the player's Shapley values in each sub-game.
- Null player: A player who contributes nothing to any coalition has a Shapley value of zero.

Owen Value Method

The Owen value is based on the idea of a "reference player" who is added to each coalition and whose contribution is distributed among the players in the coalition according to a certain rule.

- The Owen value is the result of a two-step procedure: first, the unions play a quotient game among themselves, and each one receives a payoff which, in turn, is shared among its players in an internal game.
- Both payoffs, in the quotient game for unions and within each union for its players, are given by applying the Shapley value.

Data Description

1. Dataset of **Ashraf and Galor (AER, 2013)** where the genetic diversity projections stem from, the original source: Human Genome Diversity Cell Line Panel.
2. Dataset of **Sturm and De Haan (AER, 2015)** where the dependent redistribution measure (i.e. a ratio between a Gini-coefficient before and after redistribution; the original source is: the Standardized World Income Inequality Database, SWIID) and the Economic Freedom Index (original source: Fraser Institute) come from.

108 countries over the period 1971-2010 (split up in 8 five-year intervals). The dependent variable is an average between 2001-2005.

Two datasets have been merged by **Ibrahim Ben Araar (2023)**.

23,707 observations.

Variables Description

Label	Definitions
Redistribution	Ratio between market and net Gini coefficient
Economic Freedom	5 – Year Average of Economic Freedom Index
Predicted Diversity	Predicted Heterozygosity using migratory distance to East – Africa
Ethnical Frac.	Fractionalization -ethnic
Linguistic Frac.	Fractionalization – language
Religious Frac.	Fractionalization – religion
ELF	Fractionalization – ethno – linguistic
Log GDP per Capita	Log GDP per capita (5 – year average)
KOF Globalization Index	KOF economic globalization index (Flows) (5 – year average)
Europe	Europe dummy
Asia	Asia dummy
North America	North America dummy
Oceania	Oceania
Socialist	Legal origin – socialist
Germany	Legal origin – Germany
Scandinavia	Legal origin – Scandinavia
Muslim	Share of Muslims in population (1980)
Protestant	Share of protestants in population (1980)
Catholic	Share of Catholics in population (1980)
Other Religion	Share of other religions

Descriptive Statistics

	Obs.	Mean	Std. – Dev.	Min	Max
Redistribution	4563	1.175	.210	.673	2.161
Economic Freedom	1664	6.247	1.424	2.016	9.020
Predicted Diversity	19177	.695	.059	.572	.774
Ethnical frac.	20633	.410	.255	0	.930
Linguistic frac.	20953	.349	.271	.002	.923
Religious frac.	22016	.414	.226	.002	.860
ELF	22159	.395	.299	0	.990
Log GDP per capita	8274	8.315	1.223	5.031	11.734
KOF globalization index	6786	51.568	21.885	3.140	100
Europe	19178	.289	.453	0	1
Asia	19178	.200	.400	0	1
North America	19178	.023	.151	0	1
Oceania	19178	.059	.235	0	1
Socialist	19032	.151	.358	0	1
Germany	19032	.048	.214	0	1
Scandinavia	19032	.044	.205	0	1
Muslim	19042	18.174	33.558	0	99.9
Catholic	19042	38.275	38.757	0	99.1
Protestant	18811	14.384	23.646	0	99.8
Other religions	18811	29.004	32.258	0	100

Income Redistribution, Economic Freedom, Predicted Diversity, Including Fractionalization

	(1) redistr~n	(2) redistr~n	(3) redistr~n	(4) redistr~n
ecfree	-0.295*** (-5.95)	-0.288 (-1.43)	-0.344*** (-6.20)	-0.321*** (-6.50)
pdiv	-1.869*** (-4.30)	-1.978 (-1.07)	-2.488*** (-5.15)	-1.733*** (-3.94)
c.ecfree~v	0.521*** (7.37)	0.546* (1.94)	0.625*** (7.96)	0.566*** (8.02)
frac_ade~h	-0.310*** (-14.06)			
frac_ade~n		-0.167*** (-2.63)		
frac_ade~l			-0.119*** (-4.79)	
frac_dow~f				-0.246*** (-12.86)
_cons	2.198*** (7.27)	2.018 (1.54)	2.411*** (7.14)	2.049*** (6.70)
N	943	99	943	930
R-sq	0.561	0.546	0.481	0.541

t statistics in parentheses

* p<0.10, ** p<0.05, *** p<0.01

This table shows the relationship between redistribution, economic freedom, predicted diversity and multiple measures of fractionalization.

T-statistics in parentheses.

Sources: (Sturm & De Haan, 2015a) and (Ashraf & Galor, 2013)

***Significant at the 1 percent level.

**Significant at the 5 percent level.

*Significant at the 10 percent level.

Income Redistribution and Multiple Factors

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	redistr~n	redistr~n	redistr~n	redistr~n	redistr~n	redistr~n	redistr~n	redistr~n	redistr~n	redistr~n
ecfree	-0.423*** (-7.90)	-0.361* (-1.77)	-0.314*** (-6.17)	-0.428*** (-7.80)	-0.214*** (-4.00)	-0.228*** (-5.13)	-0.361*** (-6.74)	-0.407*** (-8.38)	-0.398*** (-7.12)	-0.353*** (-6.62)
pdiv	-3.168*** (-6.78)	-3.020 (-1.63)	-1.558*** (-3.41)	-3.244*** (-6.72)	-2.473*** (-4.50)	-1.697*** (-4.40)	-2.343*** (-4.90)	-3.021*** (-7.13)	-2.865*** (-5.65)	-2.394*** (-5.09)
ecfree_p~v	0.733*** (9.64)	0.667** (2.34)	0.488*** (6.59)	0.744*** (9.44)	0.357*** (4.69)	0.418*** (6.57)	0.629*** (8.22)	0.680*** (9.84)	0.697*** (8.76)	0.632*** (8.34)
lrgdpepop			0.0944*** (13.08)							
kofgi_ec~s				-0.000338 (-0.97)						
europe					0.388*** (8.65)					
asia					0.0623* (1.66)					
africa					0.112** (2.34)					
wb_nam					0.239*** (6.67)					
oceania					0.138*** (4.51)					
legor_so						0.101*** (7.15)				
legor_ge						0.139*** (6.35)				
legor_sc						0.442*** (21.88)				
pmuslim							-0.00142*** (-6.09)			
pprotest								0.00319*** (14.29)		
pcatholic									0.000266 (1.53)	
pother										-0.00128*** (-6.84)
_cons	2.857*** (8.70)	2.611* (1.97)	1.282*** (3.85)	2.913*** (8.56)	2.524*** (6.83)	1.943*** (7.19)	2.368*** (7.13)	2.840*** (9.53)	2.637*** (7.36)	2.359*** (7.17)
N	943	100	918	928	943	943	943	935	943	935
R-sq	0.468	0.514	0.549	0.465	0.660	0.651	0.488	0.563	0.469	0.492

t statistics in parentheses
* p<0.10, ** p<0.05, *** p<0.01

The dependent variable is income redistribution.

The first column is the regular model, the second column represents the model for the year 2005.

The other columns include a variety of dummy variables in the basic model.

T – statistics in parentheses.

Source: (Ibrahim Ben Araar, 2023)

Application of Rego Model

In our work, we applied **rego method** to each OLS regression model (10 of them) in STATA to **decompose the R-squared goodness-of-fit measure** of each model into **contributions of individual regressors** and made conclusions based on the results.

	(1) redistr~n	(2) redistr~n	(3) redistr~n	(4) redistr~n	(5) redistr~n	(6) redistr~n	(7) redistr~n	(8) redistr~n	(9) redistr~n	(10) redistr~n
ecfree	-0.423*** (-7.90)	-0.361* (-1.77)	-0.314*** (-6.17)	-0.428*** (-7.80)	-0.214*** (-4.00)	-0.228*** (-5.13)	-0.361*** (-6.74)	-0.407*** (-8.38)	-0.398*** (-7.12)	-0.353*** (-6.62)
pdiv	-3.168*** (-6.78)	-3.020 (-1.63)	-1.558*** (-3.41)	-3.244*** (-6.72)	-2.473*** (-4.50)	-1.697*** (-4.40)	-2.343*** (-4.90)	-3.021*** (-7.13)	-2.865*** (-5.65)	-2.394*** (-5.09)
ecfree_p~v	0.733*** (9.64)	0.667** (2.34)	0.488*** (6.59)	0.744*** (9.44)	0.357*** (4.69)	0.418*** (6.57)	0.629*** (8.22)	0.680*** (9.84)	0.697*** (8.76)	0.632*** (8.34)
lrgdpepop			0.0944*** (13.08)							
kofgi_ec~s				-0.000338 (-0.97)						
europa					0.388*** (8.65)					
asia					0.0623* (1.66)					
africa					0.112** (2.34)					
wb_nam					0.239*** (6.67)					
oceania					0.138*** (4.51)					
legor_so						0.101*** (7.15)				
legor_ge						0.139*** (6.35)				
legor_sc						0.442*** (21.88)				
pmuslim							-0.00142*** (-6.09)			
pprotest								0.00319*** (14.29)		
pcatholic									0.000266 (1.53)	
pother										-0.00128*** (-6.84)
_cons	2.857*** (8.70)	2.611* (1.97)	1.282*** (3.85)	2.913*** (8.56)	2.524*** (6.83)	1.943*** (7.19)	2.368*** (7.13)	2.840*** (9.53)	2.637*** (7.36)	2.359*** (7.17)
N	943	100	918	928	943	943	943	935	943	935
R-sq	0.468	0.514	0.549	0.465	0.660	0.651	0.488	0.563	0.469	0.492

t statistics in parentheses
* p<0.10, ** p<0.05, *** p<0.01

Outcomes

```
. rego redistribution ecfree pdiv ecfree_pdiv
```

Gr	Regressor	Coef.	Std.Err.	P> t	Std.Coeff.	Shapley %R2
1	ecfree	-.4228996 ***	.0535461	0.000	-2.5533	37.4295
2	pdiv	-3.168471 ***	.4670933	0.000	-0.7631	13.0194
3	ecfree_pdiv	.7330411 ***	.0760327	0.000	3.3500	49.5510
-	Intercept	2.8573	.3284286	0.000		
Observations		943				
Overall R2		0.46811				
Root MSE		.1716714				
F-stat. Model		275.4713 ***		0.000		
Log Likelihood		325.6744				

```
.
. rego redistribution ecfree pdiv ecfree_pdiv if year == 2005
```

Gr	Regressor	Coef.	Std.Err.	P> t	Std.Coeff.	Shapley %R2
1	ecfree	-.3608875 *	.203991	0.080	-2.0337	40.4229
2	pdiv	-3.019667	1.852362	0.106	-0.7241	8.3435
3	ecfree_pdiv	.6667605 **	.2853539	0.022	2.8191	51.2336
-	Intercept	2.61055 *	1.326244	0.052		
Observations		100				
Overall R2		0.51422				
Root MSE		.1651143				
F-stat. Model		33.87409 ***		0.000		
Log Likelihood		40.25899				

Rego method depicted that:

- ~50% of the variation in the dependent variable is explained by the **interaction between EF and predicted diversity** in the “basic” model.
- **Economic Freedom Index** has a strong descriptive value as well, while **predicted diversity** is not very (comparatively) descriptive.

This pattern we can see in other regressions as well.

In all models, R^2 varies from 46% to 66%.

```
. rego redistribution ecfree pdiv ecfree_pdiv lrgdpepop
```

Gr	Regressor	Coef.	Std.Err.	P> t	Std.Coeff.	Shapley %R2
1	ecfree	-.3144474 ***	.0510018	0.000	-1.8820	22.4941
2	pdiv	-1.557558 ***	.4573758	0.001	-0.3735	11.0327
3	ecfree_pdiv	.4875497 ***	.0739508	0.000	2.2082	30.7080
4	lrgdpepop	.0943558 ***	.0072138	0.000	0.4913	35.7651
-	Intercept	1.281686	.3325471	0.000		
Observations		918				
Overall R2		0.54898				
Root MSE		.1585782				
F-stat. Model		277.8281 ***		0.000		
Log Likelihood		390.4252				

Log GDP per capita (5 – year average) has the biggest share in R² measure (~35.8%) in presented regression.

```
.
. rego redistribution ecfree pdiv ecfree_pdiv kofgi_ecoflows
```

Gr	Regressor	Coef.	Std.Err.	P> t	Std.Coeff.	Shapley %R2
1	ecfree	-.4280454 ***	.0549081	0.000	-2.5783	32.9243
2	pdiv	-3.243505 ***	.4830141	0.000	-0.7828	12.7914
3	ecfree_pdiv	.7444594 ***	.0788211	0.000	3.3971	44.9109
4	kofgi_ecoflows	-.0003385	.0003505	0.334	-0.0297	9.3734
-	Intercept	2.913054	.3401836	0.000		
Observations		928				
Overall R2		0.46535				
Root MSE		.1722121				
F-stat. Model		200.8434 ***		0.000		
Log Likelihood		318.1103				

KOF economic globalization index (Flows) (5 – year average) has comparatively low descriptive value (~9.3% of R²).


```
. rego redistribution ecfree pdiv ecfree_pdiv europe asia africa wb_nam oceania
```

Gr	Regressor	Coef.	Std.Err.	P> t	Std.Coeff.	Shapley %R2
1	ecfree	-.2139279 ***	.0535148	0.000	-1.2916	16.2121
2	pdiv	-2.472675 ***	.5498522	0.000	-0.5955	7.8347
3	ecfree_pdiv	.3572763 ***	.076222	0.000	1.6327	21.5049
4	europe	.3876558 ***	.0447977	0.000	0.8009	37.5349
5	asia	.062319 *	.0375592	0.097	0.1006	7.2245
6	africa	.1118312 **	.0477413	0.019	0.1855	7.6483
7	wb_nam	.2389911 ***	.0358268	0.000	0.1666	1.4217
8	oceania	.1377527 ***	.0305718	0.000	0.1109	0.6188
-	Intercept	2.524395	.3694766	0.000		
Observations		943				
Overall R2		0.66009				
Root MSE		.1376042				
F-stat. Model		226.7203 ***		0.000		
Log Likelihood		536.7842				

The **North America dummy** and **Oceania dummy** have low explanatory percentage (~1.4% and ~0.6%, respectively), while **Europe dummy** has a considerable contribution (~37.5% of R^2).

```
.
. rego redistribution ecfree pdiv ecfree_pdiv legor_so legor_ge legor_sc
```

Gr	Regressor	Coef.	Std.Err.	P> t	Std.Coeff.	Shapley %R2
1	ecfree	-.228333 ***	.0445102	0.000	-1.3786	20.9818
2	pdiv	-1.697206 ***	.3860145	0.000	-0.4088	6.9298
3	ecfree_pdiv	.4182109 ***	.0636514	0.000	1.9112	27.7008
4	legor_so	.1014068 ***	.0141747	0.000	0.1464	2.8777
5	legor_ge	.1387486 ***	.0218407	0.000	0.1311	3.7233
6	legor_sc	.4417192 ***	.0201912	0.000	0.4661	37.7866
-	Intercept	1.943163	.2702632	0.000		
Observations		943				
Overall R2		0.65123				
Root MSE		.1392361				
F-stat. Model		291.2883 ***		0.000		
Log Likelihood		524.658				

Legal origin – Scandinavia explains a big share of volatility of redistribution (~37.7%), while **Legal origin – socialist** and **Legal origin – Germany** do not have a strong descriptive value (2.9% and 3.7%, respectively).

```
. rego redistribution ecfree pdiv ecfree_pdiv pmuslim
```

Gr	Regressor	Coef.	Std.Err.	P> t	Std.Coeff.	Shapley %R2
1	ecfree	-.3608941 ***	.0535202	0.000	-2.1790	32.0937
2	pdiv	-2.342508 ***	.4779709	0.000	-0.5642	13.8171
3	ecfree_pdiv	.6289909 ***	.0765387	0.000	2.8745	43.5810
4	pmuslim	-.0014196 ***	.0002329	0.000	-0.1666	10.5083
-	Intercept	2.367924	.3321368	0.000		
Observations		943				
Overall R2		0.48837				
Root MSE		.1684599				
F-stat. Model		223.842 ***		0.000		
Log Likelihood		343.9851				

Share of Muslims in population (1980) contributes 10.5% to the description of the variation in the redistribution.

```
.
. rego redistribution ecfree pdiv ecfree_pdiv pprotest
```

Gr	Regressor	Coef.	Std.Err.	P> t	Std.Coeff.	Shapley %R2
1	ecfree	-.4071476 ***	.0486055	0.000	-2.4671	26.1296
2	pdiv	-3.020907 ***	.4239532	0.000	-0.7292	9.6022
3	ecfree_pdiv	.6797797 ***	.0690943	0.000	3.1160	34.9771
4	pprotest	.0031901 ***	.0002232	0.000	0.3395	29.2912
-	Intercept	2.84012	.2980176	0.000		
Observations		935				
Overall R2		0.56292				
Root MSE		.1557602				
F-stat. Model		299.4432 ***		0.000		
Log Likelihood		414.3731				

Share of protestants in population (1980) has a considerate weight in explanation of income inequality – almost 30% of R^2.

```
. rego redistribution ecfree pdiv ecfree_pdiv patholic
```

Gr	Regressor	Coef.	Std.Err.	P> t	Std.Coeff.	Shapley %R2
1	ecfree	-.3980978 ***	.0559223	0.000	-2.4036	36.9664
2	pdiv	-2.865214 ***	.5073096	0.000	-0.6901	13.1673
3	ecfree_pdiv	.6970473 ***	.0795562	0.000	3.1855	48.8103
4	patholic	.0002663	.0001745	0.127	0.0434	1.0559
-	Intercept	2.637064	.3585293	0.000		
Observations		943				
Overall R2		0.46943				
Root MSE		.1715501				
F-stat. Model		207.4779 ***		0.000		
Log Likelihood		326.8434				

```
.
. rego redistribution ecfree pdiv ecfree_pdiv pother
```

Gr	Regressor	Coef.	Std.Err.	P> t	Std.Coeff.	Shapley %R2
1	ecfree	-.3526227 ***	.0532923	0.000	-2.1367	35.0659
2	pdiv	-2.394206 ***	.4701571	0.000	-0.5779	12.6968
3	ecfree_pdiv	.6315036 ***	.0757173	0.000	2.8947	46.6664
4	pother	-.0012798 ***	.000187	0.000	-0.1684	5.5708
-	Intercept	2.359439	.3290238	0.000		
Observations		935				
Overall R2		0.49249				
Root MSE		.1678417				
F-stat. Model		225.6197 ***		0.000		
Log Likelihood		344.5257				

Share of Catholics in population (1980) plays less significant role in income inequality explanation and contributes only 1% to R^2 .

Same we can witness with **Share of other religions** variable. Its share in R^2 is only 5.6%.

Conclusion

We made an extended analysis of Ibrahim Ben Araar's work (based on Sturm & De Haan (2015) and Ashraf & Galor (2013) researches) using rego method - decomposed the R-squared goodness-of-fit measure of OLS regression models into contributions of individual regressors.

Results:

1. The **interaction between EF and predicted diversity** explain a large proportion of the variation in the dependent variable (redistribution – a ratio between market and net Gini coefficient) in every regression (from 21% to 51%).
2. The **Economic Freedom Index** has a strong explanation value as its decomposed share of R^2 is usually $>20\%$ in every regression.
3. The **predicted diversity** usually contribute less to the explanation of dependent variable (from $\sim 6.9\%$ of R^2 to $\sim 13.8\%$).

Conclusion

4. The **Log GDP per capita** (5 – year average) contributes considerable share to the explanatory effectiveness of the OLS model (~35.8%).
5. The **North America dummy** and **Oceania dummy** have low explanatory percentage (~1.4% and ~0.6%, respectively), while **Europe dummy** has a considerable contribution (~37.5% of R^2).
6. The **Legal origin – Scandinavia** explains a big share of volatility of redistribution (~37.7%), while **Legal origin – socialist** and **Legal origin – Germany** do not have a strong descriptive value (2.9% and 3.7%, respectively).
7. The **Share of protestants in population** (1980) plays an important role in explaining the variations in redistribution (almost 30% of R^2), while **Share of Muslims in population** contributes 10.5%, and **Share of other religions** and **Share of Catholics in population** make a slight contribution to goodness-of-fit measure (5.6% and 1%, respectively).

Thank you for your
attention!